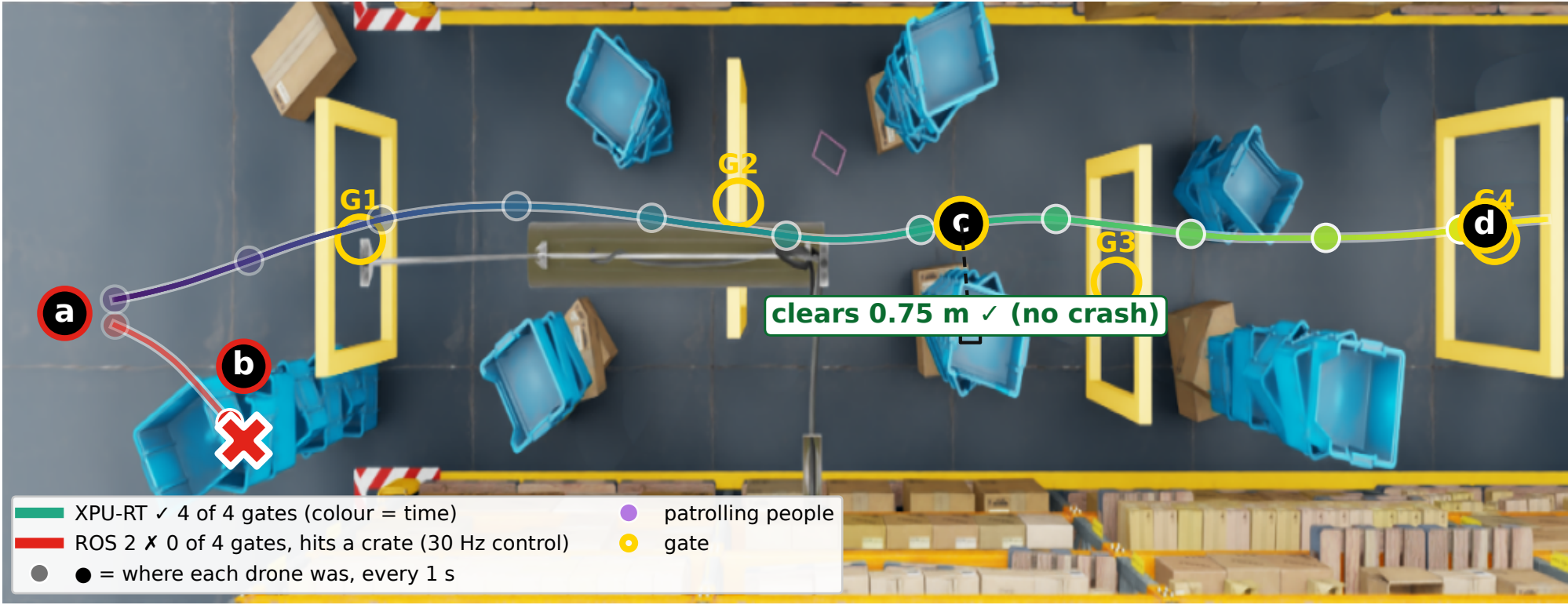
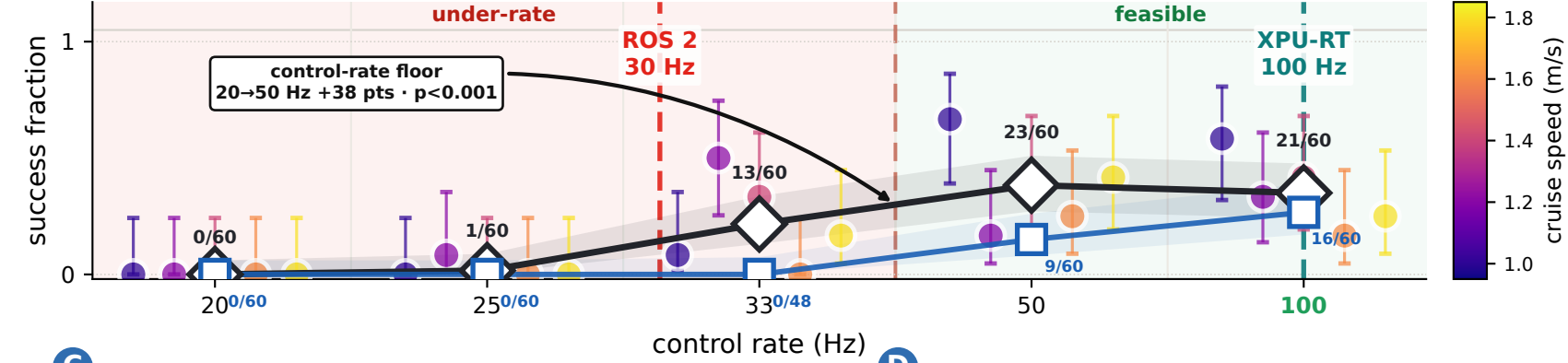


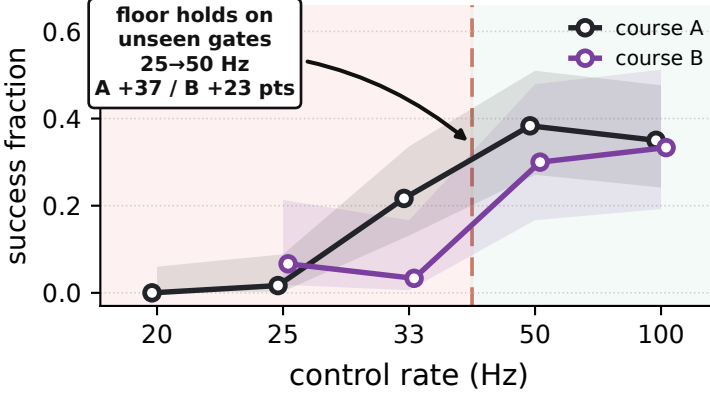
Warehouse gate-course showdown — same scene, drone and gain, at 1.4 m/s
XPU-RT · solver-placed CP-SAT schedule, conv on the IME (100 Hz control) clears all 4 gates
ROS 2 · two YOLO pools over all eight harts, unpinned, control chained to the goal (30 Hz control) hits a crate
one seed of this scene (seed 1003); over its 12 seeds XPU-RT completes 3, ROS 2 none



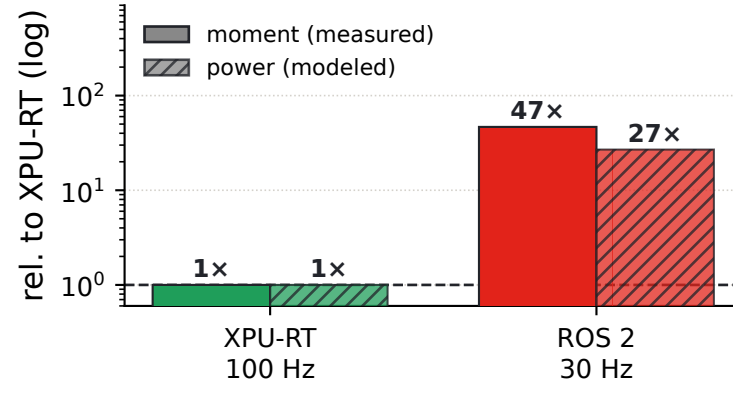
B Control-rate envelope (rate injected, 300 flights): success against control rate — ROS 2's 30 Hz and XPU-RT's 100 Hz marked



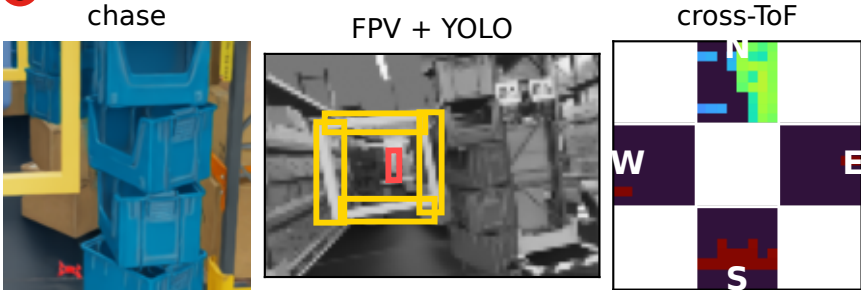
C Generalization: unseen gates (course B)



D Mechanism: the baseline thrashes (n=6 flights/arm)



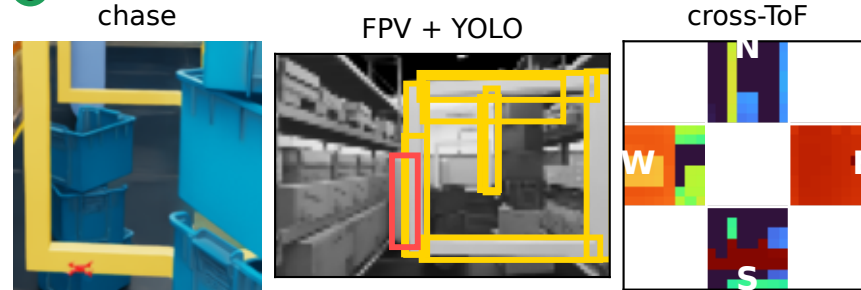
a ROS 2 30 Hz · has not reached G1 · t=0.5s



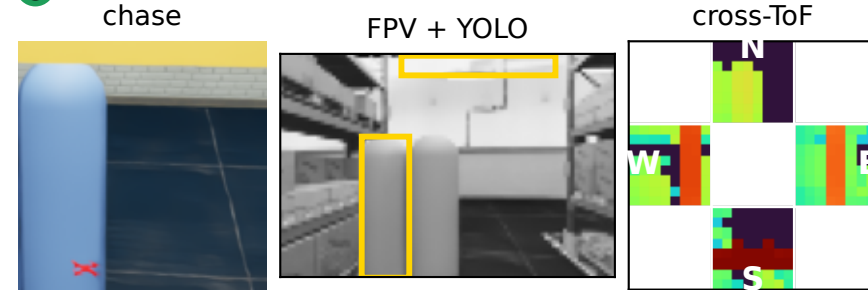
b ROS 2 30 Hz · hits crate before G1 · t=2.0s



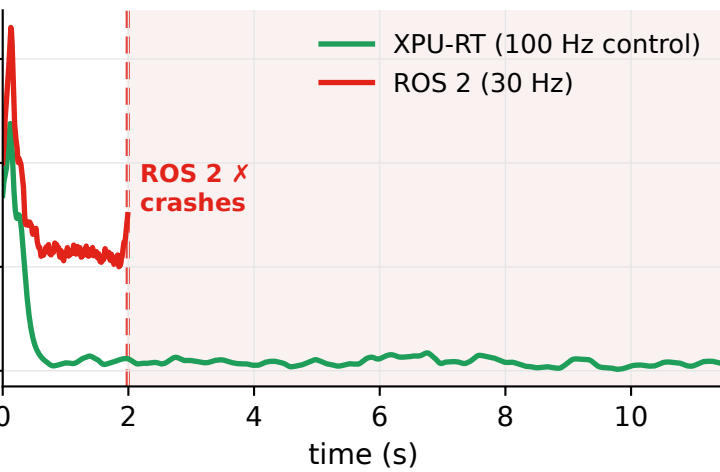
c XPU-RT 100 Hz · clears crate 0.75 m · t=7.2s



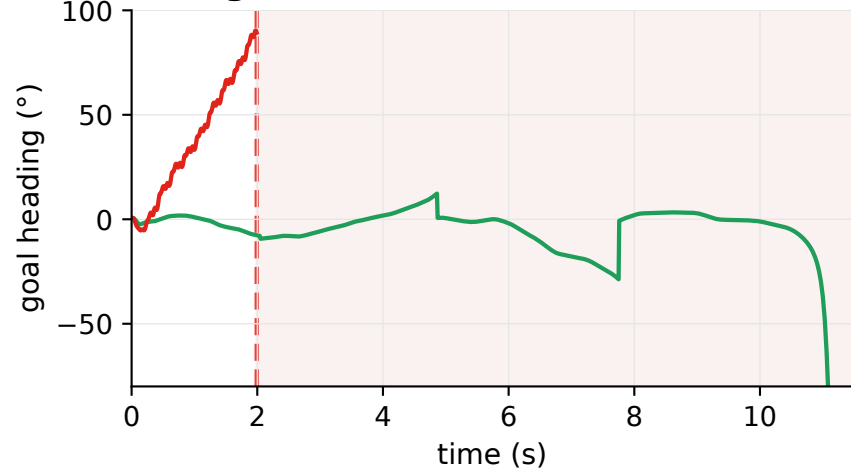
d XPU-RT 100 Hz · reaches gate · t=11.0s



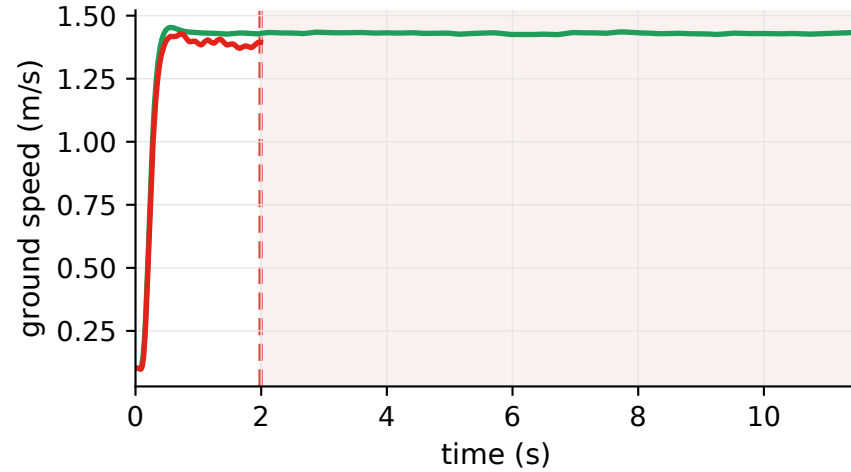
E Inside the pair — body rate: the baseline thrashes at 30 Hz



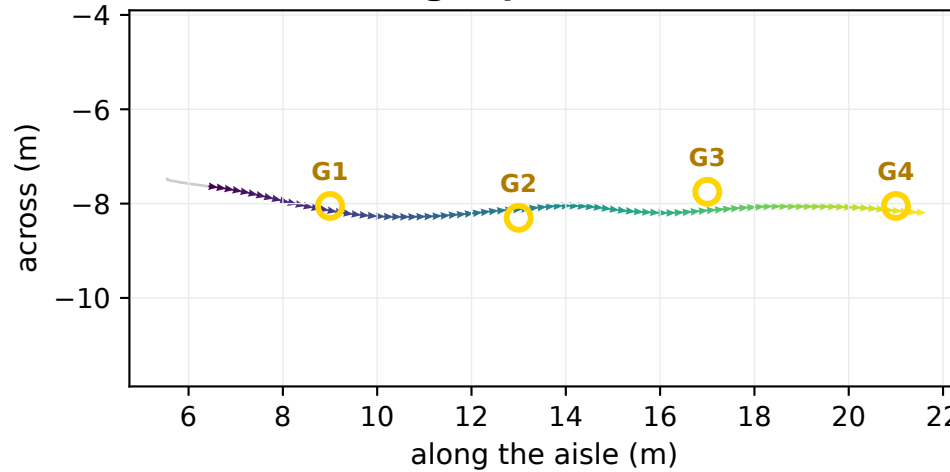
F Nav goal heading: the two commands diverge before the baseline's collision



G Forward speed: the same cruise until the baseline's collision



H XPU-RT's velocity through the gates (arrow = heading · speed, colour = time)



I Onboard K1 schedule, measured at the 30 Hz camera — XPU-RT · solver-placed CP-SAT schedule, conv on the IME places the kernels over all 8 harts: camera→control 27 ms, control every 10.0 ms, every frame on time · YOLO on 8 harts, 5 traced, nav on 5 harts, 2 traced, control on 1 hart
ROS 2 · all 8 cores (5 processes · single executor · chained control · unpinned): camera→control 31 ms, control every 33.3 ms, 0/298 frames late · YOLO on 4 harts, control on 1 hart, nav on 1 hart

